2.1.32

$$\begin{cases} x_1 - x_2 = 1 \\ 2x_1 - 2x_2 = 5 \end{cases}$$

□ $(A|B) = \left(\begin{array}{cc|c} 1 & -1 & 1 \\ 2 & -2 & 5 \end{array}\right)_{II-2\cdot I} \sim \left(\begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 0 & 3 \end{array}\right)$

$r(A)=1$, $r(A|B)=2$ $\Rightarrow r(A) < r(A|B) \Rightarrow$ система несовместима $\Rightarrow$ нет решений.

Ответ: нет решений. ■

2.1.33

$$\begin{cases} 3x + 2y = 5 \\ 6x + 4y = 10 \end{cases}$$

□ $(A|B) = \left(\begin{array}{cc|c} 3 & 2 & 5 \\ 6 & 4 & 10 \end{array}\right)_{II-2\cdot I} \sim \left(\begin{array}{cc|c} 3 & 2 & 5 \\ 0 & 0 & 0 \end{array}\right)$

$r(A) = r(A|B) = 1 \Rightarrow$ система совместима

$n = 2 \Rightarrow r < n$

$r = 1 \Rightarrow 1$ гл пер-ная

$n - r = 1 \Rightarrow 1$ св пер-ая

$|a_{11}| = |3| = 3 \Rightarrow$ x гл пер, y св пер

$3x+2y=5$

$3x=5-2y$

$x=\dfrac{5-2y}{3}$

$\sqsupset$ $y=t \Rightarrow x=\dfrac{5-2t}{3}$

Общ. решение $\left(\dfrac{5-2t}{3};t\right)$

$\sqsupset$ $t=2, \Rightarrow \left(\dfrac{1}{3};2\right)$ частное решение

Ответ: $\left(\dfrac{5-2t}{3};t\right)$ - общее

$\left(\dfrac{1}{3};2\right)$ - частное.

2.1.34

$$\begin{cases} x_1+2x_2=3 \\ -2x_1+3x_2=0 \\ -2x_1-4x_2=1 \end{cases}$$

$\square$

$$(A|B)=\left(\begin{array}{cc|c} 1 & 2 & 3 \\ -2 & 3 & 0 \\ -2 & -4 & 1 \end{array}\right) \begin{array}{l} \\ II+2\cdot I \\ III\cdot 2I \end{array} \sim \left(\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 7 & 6 \\ 0 & 0 & 7 \end{array}\right)$$

$r(A)=2$

$r(A|B)=3$ $\Big|\Rightarrow$ нет решений

Ответ: нет решений.

2.1.35

$$\begin{cases} x-\sqrt{3}y=1 \\ \sqrt{3}x-3y=\sqrt{3} \\ -\frac{\sqrt{3}}{3}x+y=-\frac{\sqrt{3}}{3} \end{cases}$$

$\square$ $(A|B)=\left(\begin{array}{cc|c} 1 & -\sqrt{3} & 1 \\ \sqrt{3} & -3 & \sqrt{3} \\ -\frac{\sqrt{3}}{3} & 1 & -\frac{\sqrt{3}}{3} \end{array}\right)$ $\begin{array}{l} II-\sqrt{3}\cdot I \\ \sqrt{3}\cdot III+I \end{array}$ $\sim\left(\begin{array}{cc|c} 1 & -\sqrt{3} & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right)$

$r(A)=r(A|B)=1 \Rightarrow$ совм

$n=2 \Rightarrow r<n$

$n-r=1$

$|\Delta_1|=|1|=1\neq 0$ $\quad$ x – ил, y – св

$x-\sqrt{3}y=1$

$x=1+\sqrt{3}y$

] $y=\sqrt{3}t \Rightarrow x=1+3t \Rightarrow$ Общ реш $=(1+3t;\sqrt{3}t)$

] $t=1 \Rightarrow$ част реш $(4;\sqrt{3})$

Ответ: Общ реш $(1+3t;\sqrt{3}t)$

част реш $(4;\sqrt{3})$ $\blacksquare$

~~[illegible]~~

2.1.36

$$\begin{cases} 3x - y = -5 \\ 2x + 3y = 4 \\ -x + \frac{1}{3}y = \frac{5}{3} \\ x + 1{,}5y = 2 \end{cases}$$

$\square$

$$(A|B) = \left(\begin{array}{cc|c} 3 & -1 & -5 \\ 2 & 3 & 4 \\ -1 & \frac{1}{3} & \frac{5}{3} \\ 1 & 1{,}5 & 2 \end{array}\right) \begin{array}{l} \\ II \cdot 3 - 2 \cdot I \\ III + \frac{1}{3} \cdot I \\ IV - \frac{1}{3} \cdot I \end{array} \sim \left(\begin{array}{cc|c} 3 & -1 & -5 \\ 0 & 11 & 22 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right)$$

$r(A) = r(A|B) = 2 \Rightarrow$ совм

$n = 2 \Rightarrow r = n$

$$\begin{cases} 3x - y = -5 \\ 11y = 22 \end{cases} \quad \begin{cases} 3x - y = -5 \\ y = 2 \end{cases} \quad \begin{cases} x = -1 \\ y = 2 \end{cases}$$

одн. и т.ч. реш $(-1; 2)$

Ответ: одн. и т.ч. р-е $(-1; 2)$ ■

2.1.37

$$\begin{cases} 3x+4y+2z=8 \\ 2x-4y-3z=-1 \\ x+5y+z=0 \end{cases}$$

$$(A|B)=\left(\begin{array}{ccc|c} 3 & 4 & 2 & 8 \\ 2 & -4 & -3 & -1 \\ 1 & 5 & 1 & 0 \end{array}\right)\begin{array}{l} \\ II\cdot 3-I\cdot 2 \\ III\cdot 3-I \end{array} \sim$$

$$\sim\left(\begin{array}{ccc|c} 3 & 4 & 2 & 8 \\ 0 & -20 & -13 & -19 \\ 0 & 11 & 1 & -8 \end{array}\right)\begin{array}{l} \\ \\ III\cdot 20+II\cdot 11 \end{array} \sim \left(\begin{array}{ccc|c} 3 & 4 & 2 & 8 \\ 0 & -20 & -13 & -19 \\ 0 & 0 & -123 & -369 \end{array}\right)$$

$r(A)=r(A|B)=3 \Rightarrow$ система совм.

$n=r \Rightarrow$

$$\begin{cases} 3x+4y+2z=8 \\ -20y-13z=-19 \\ -123\cdot z=-369 \end{cases} \Leftrightarrow \begin{cases} x=2 \\ y=-1 \\ z=3 \end{cases}$$

Общее и частное решение $=(2;-1;3)$

Ответ: Общее и частное решение $(2;-1;3)$

2.1.38

$$\begin{cases} -x + y - 3z = 5 \\ 3x - y - z = 2 \\ 2x + y - 9z = 0 \end{cases}$$

□ $(A|B) = \left(\begin{array}{ccc|c} -1 & 1 & -3 & 5 \\ 3 & -1 & -1 & 2 \\ 2 & 1 & -9 & 0 \end{array}\right) \begin{array}{l} \\ II + I \cdot 3 \\ III + I \cdot 2 \end{array} \sim$

$\sim \left(\begin{array}{ccc|c} -1 & 1 & -3 & 5 \\ 0 & 2 & -10 & 17 \\ 0 & 3 & -19 & 10 \end{array}\right) \begin{array}{l} \\ \\ III \cdot 2 - II \cdot 3 \end{array} \sim \left(\begin{array}{ccc|c} -1 & 1 & -3 & 5 \\ 0 & 2 & -10 & 17 \\ 0 & 0 & 0 & -3\dots \end{array}\right)$

~~[illegible]~~ $r(A|B) > r(A) \Rightarrow$ ~~[illegible]~~ несовместна $\Rightarrow$

$\Rightarrow$ нет решения

Ответ: нет решения

2.1.41

$$\begin{cases} 3x_1 + 2x_2 + x_3 = 5 \\ 2x_1 + 3x_2 + x_3 = 1 \\ 2x_1 + x_2 + 3x_3 = 11 \\ 3x_1 + 4x_2 - x_3 = -9 \end{cases}$$

□

$$(A|B) = \left(\begin{array}{ccc|c} 3 & 2 & 1 & 5 \\ 2 & 3 & 1 & 1 \\ 2 & 1 & 3 & 11 \\ 3 & 4 & -1 & -5 \end{array}\right) \begin{array}{l} \\ II\cdot 3 - I\cdot 2 \\ III\cdot 3 - I\cdot 2 \\ IV - I \end{array} \sim$$

$$\sim \left(\begin{array}{ccc|c} 3 & 2 & 1 & 5 \\ 0 & 5 & 1 & -7 \\ 0 & -1 & 7 & 23 \\ 0 & 2 & -2 & -10 \end{array}\right) \begin{array}{l} \\ \\ III\cdot 5 + II \\ IV\cdot 5 - II\cdot 2 \end{array} \sim \left(\begin{array}{ccc|c} 3 & 2 & 1 & 5 \\ 0 & 5 & 1 & -7 \\ 0 & 0 & 36 & 108 \\ 0 & 0 & -12 & -36 \end{array}\right) \begin{array}{l} \\ \\ \\ 3\cdot IV + III \end{array} \sim$$

$$\sim \left(\begin{array}{ccc|c} 3 & 2 & 1 & 5 \\ 0 & 5 & 1 & -7 \\ 0 & 0 & 36 & 108 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$r(A) = r(A|B) = 3 \Rightarrow$ Совмест

$r = n = 3$

$$\begin{cases} 3x_1 + 2x_2 + x_3 = 5 \\ 5x_2 + x_3 = -7 \\ 36x_3 = 108 \end{cases} \qquad \begin{cases} x_1 = 2 \\ x_2 = -2 \\ x_3 = 3 \end{cases}$$

Од ч част решен $(2; -2; 3)$

Ответ: Од ч частн реш $(2; -2; 3)$

2.1.47

$$\begin{cases} 2\sqrt{5}\,x_1 - x_2 + \sqrt{5}\,x_3 = 1 \\ 10x_1 - \sqrt{5}\,x_2 + 5x_3 = \sqrt{5} \\ -2x_1 + \frac{\sqrt{5}}{5}x_2 - x_3 = -\frac{1}{\sqrt{5}} \end{cases}$$

□

$$(A|B) = \left(\begin{array}{ccc|c} 2\sqrt{5} & -1 & \sqrt{5} & 1 \\ 10 & -\sqrt{5} & 5 & \sqrt{5} \\ -2 & \frac{\sqrt{5}}{5} & -1 & -\frac{1}{\sqrt{5}} \end{array}\right) \begin{array}{l} \\ II - \sqrt{5}\cdot I \\ III + \frac{1}{\sqrt{5}}\cdot I \end{array} \sim$$

$$\sim \left(\begin{array}{ccc|c} 2\sqrt{5} & -1 & \sqrt{5} & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$r(A) = r(A|B) = 1 \Rightarrow$

$n = 3 \Rightarrow \infty$ rj

$3 - 1 = 2.$

$|a_{11}| = |2\sqrt{5}| = 2\sqrt{5} \neq 0$

x_1 - zn

x_2, x_3 - ln

$2\sqrt{5}\,x_1 - x_2 + \sqrt{5}\,x_3 = 1$

$$x_1 = \frac{x_2 - \sqrt{5}\,x_3 + 1}{2\sqrt{5}}$$

$\exists\ x_2 = t_1$

$x_3 = t_2$

$$Oo\delta_3 = \left(\frac{t_1 - \sqrt{5}\,t_2 + 1}{2\sqrt{5}}; t_1; t_2\right)$$

$\exists\ t_1 = 0$
$t_2 = 1$ час $\left(\frac{-5+4\sqrt{5}}{10}; 0; 1\right)$

Ответ: $\text{част} = \left(\frac{t_1 - \sqrt{5}t_2 + 1}{2\sqrt{5}}; t_1; t_2\right)$

$$Oo\delta_3 = \left(\frac{-5+4\sqrt{5}}{10}; 0; 1\right)$$

<u>2.7.43</u>

$$\begin{cases} 3x_1 + 4x_2 + x_3 + 2x_4 = 3 \\ 6x_1 + 8x_2 + 2x_3 + 5x_4 = 7 \\ 9x_1 + 12x_2 + 3x_3 + 10x_4 = 13 \end{cases}$$

□

$$(A|B) = \left(\begin{array}{cccc|c} 3 & 4 & 1 & 2 & 3 \\ 6 & 8 & 2 & 5 & 7 \\ 9 & 12 & 3 & 10 & 13 \end{array}\right) \begin{array}{l} \\ II - 2\cdot I \\ III - 3\cdot I \end{array} \sim \sim \left(\begin{array}{cccc|c} 3 & 4 & 1 & 2 & 3 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 4 & 4 \end{array}\right) \underset{III - 4\cdot II}{\sim}$$

$$\sim \left(\begin{array}{cccc|c} 3 & 4 & 1 & 2 & 3 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$r(A) = r(A|B) = 2 \Rightarrow$ совм

$n - r = 2 \Rightarrow$ св пер

$$\begin{vmatrix} a_{13} & a_{14} \\ a_{23} & a_{24} \end{vmatrix} = \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} = 1\cdot 1 - 2\cdot 0 = 1 \neq 0$$

$\Rightarrow x_3, x_4$ – бз

x_1, x_2 св

$$\begin{cases} 3x_1 + 4x_2 + x_3 + 2x_4 = 3 \\ x_4 = 1 \end{cases} \quad \begin{cases} x_3 = 1 - 3x_1 - 4x_2 \\ x_4 = 1 \end{cases}$$

$$\text{П } x_1 = t_2 \quad \begin{cases} x_3 = 1 - 3t_1 - 4t_2 \\ x_4 = 1 \end{cases}$$
$$x_2 = t_2$$

Общее

$(t_1; t_2; 1 - 3t_1 - 4t_2; 1)$

П $t_1 = 1$
$t_2 = 0$ $\Rightarrow (1; 0; -2; 1)$ – част

Ответ: Общ $(t_1; t_2; 1 - 3t_1 - 4t_2; 1)$
част $(1; 0; -2; 1)$

2.1.46

$$\begin{cases} 6x_1+4x_2+5x_3+2x_4+3x_5=1 \\ 3x_1+2x_2+4x_3+x_4+2x_5=3 \\ 3x_1+2x_2-2x_3+x_4=-7 \\ 9x_1+6x_2+x_3+3x_4+2x_5=2 \end{cases}$$

$\square$

$$(A|B)=\left(\begin{array}{ccccc|c} 6&4&5&2&3&1 \\ 3&2&4&1&2&3 \\ 3&2&-2&1&0&-7 \\ 9&6&1&3&2&2 \end{array}\right) \begin{array}{l} \\ 2\cdot II - I \\ 2\cdot III - I \\ 2\cdot IV - 3\cdot I \end{array} \sim \left(\begin{array}{ccccc|c} 6&4&5&2&3&1 \\ 0&0&3&0&1&5 \\ 0&0&-9&0&-3&-15 \\ 0&0&-13&0&-5&1 \end{array}\right) \begin{array}{l} \\ \\ III+3\cdot II \\ 1\cdot IV+13\cdot II \end{array} \sim$$

$$\sim \left(\begin{array}{ccccc|c} 6&4&5&2&3&1 \\ 0&0&3&0&1&5 \\ 0&0&0&0&0&0 \\ 0&0&0&0&-2&68 \end{array}\right) \begin{array}{l} \\ \\ III \leftrightarrow IV \\ \\ \end{array} \sim \left(\begin{array}{ccccc|c} 6&4&5&2&3&1 \\ 0&0&3&0&1&5 \\ 0&0&0&0&-2&68 \\ 0&0&0&0&0&0 \end{array}\right)$$

$r(A) = r(A|B) = 3 \Rightarrow$ совмест

$n(A) = 3$ гл

$n - r = 5 - 3 = 2$ св

$$\begin{vmatrix} a_{13} & a_{14} & a_{15} \\ a_{23} & a_{24} & a_{25} \\ a_{33} & a_{34} & a_{35} \end{vmatrix} = \begin{vmatrix} 5 & 2 & 3 \\ 3 & 0 & 1 \\ 0 & 0 & -2 \end{vmatrix} = 12 \quad \det = 12 \neq 0 \Rightarrow$$

$x_3; x_4; x_5$ – гл

$x_1; x_2$ – св

$$\begin{cases} 6x_1+4x_2+5x_3+2x_4+3x_5=1 \\ 3x_3+x_5=5 \\ -2x_5=68 \end{cases} \qquad \begin{cases} x_4=19-2x_2-3x_1 \\ x_3=13 \\ x_5=-34 \end{cases}$$

Ɔ $x_1=t_1$ $\qquad x_4=19-2t_2-3\cdot t_1$

$x_2=t_2$

Общее $(t_1;t_2;13;19-2t_2-3t_1;-34)$

Ɔ $t_1=0$

$t_2=1$

Частное $(0;1;13;17;-34)$

Ответ: Общ. $(t_1;t_2;13;19-2t_2-3t_1;-34)$

част. $(0;1;13;17;-34)$

<u>2.1.47</u>

$x_1+x_2+3x_3-2x_4+3x_5=1$

$2x_1+2x_2+4x_3-x_4+3x_5=2$

$3x_1+3x_2+5x_3-2x_4+3x_5=1$

$2x_1+2x_2+8x_3-3x_4+0x_5=2$

□

$$(A|B)=\left(\begin{array}{ccccc|c} 1 & 1 & 3 & -2 & 3 & 1 \\ 2 & 2 & 4 & -1 & 3 & 2 \\ 3 & 3 & 5 & -2 & 3 & 1 \\ 2 & 2 & 8 & -3 & 0 & 2 \end{array}\right) \begin{array}{l} \\ II-2I \\ III-3I \\ IV-2I \end{array} \sim \left(\begin{array}{ccccc|c} 1 & 1 & 3 & -2 & 3 & 1 \\ 0 & 0 & -2 & 3 & -3 & 0 \\ 0 & 0 & -4 & 4 & -6 & -2 \\ 0 & 0 & 2 & 1 & 3 & 0 \end{array}\right)$$

$$\sim \left(\begin{array}{ccccc|c} 1 & 1 & 3 & -2 & 3 & 1 \\ 0 & 0 & -2 & 3 & -3 & 0 \\ 0 & 0 & 0 & -2 & 0 & -2 \\ 0 & 0 & 0 & 4 & 0 & 0 \end{array}\right) \underset{IV+2\cdot III}{\sim} \left(\begin{array}{ccccc|c} 1 & 1 & 3 & -2 & 3 & 1 \\ 0 & 0 & -2 & 3 & -3 & 0 \\ 0 & 0 & 0 & -2 & 0 & -2 \\ 0 & 0 & 0 & 0 & 0 & -4 \end{array}\right)$$

$r(A) = 3 < r(A|\bar{B}) \Rightarrow$ нет решений

Ответ: нет решений ■

2.2.17

$f(x) = a \cdot 3^x + b \cdot x^2 + c$

□ $f(0) = 2 \cdot a + 0 \cdot b + c = 2$

$a + c = 2$

$f(1) = -1 : 3a + b + c = -1$

$f(2) = 4 : 9a + 4b + c = 4$

$$\begin{cases} a + c = 2 \\ 3a + b + c = -1 \\ 9a + 4b + c = 4 \end{cases}$$

1) $AX = B \Rightarrow X = A^{-1} \cdot B$

$A = \begin{pmatrix} 1 & 0 & 1 \\ 3 & 1 & 1 \\ 9 & 4 & 1 \end{pmatrix}$ $\det(A) = 0 \Rightarrow \nexists A^{-1} \Rightarrow$ ~~т~~ определитель равен 0 $\Rightarrow$ нельзя решить

Ответ: нельзя решить. ■

2.2.18

$$\begin{cases} x_1 - x_2 = 5 \\ 2x_1 + x_2 = 1 \end{cases}$$

$\square$

1) $AX = B \Rightarrow X = A^{-1} \cdot B$

$$A = \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}$$

$$\det(A) = \begin{vmatrix} 1 & -1 \\ 2 & 1 \end{vmatrix} = 1 \cdot 1 - (-1)2 = 3 \neq 0 \Rightarrow \exists A^{-1}$$

$$A^{-1} = \frac{1}{\det(A)} \cdot \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 1 & 1 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} \\ -\frac{2}{3} & \frac{1}{3} \end{pmatrix}$$

$$X = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} \\ -\frac{2}{3} & \frac{1}{3} \end{pmatrix} \cdot \begin{pmatrix} 5 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{3} \cdot 5 + \frac{1}{3} \cdot 1 \\ (-\frac{2}{3}) \cdot 5 + \frac{1}{3} \end{pmatrix} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$$

2) $D = \det(A) = 3$

$$D_1 = \begin{vmatrix} 5 & -1 \\ 1 & 1 \end{vmatrix} = 5 \cdot 1 - (-1) \cdot 1 = 6$$

$$x_1 = \frac{D_1}{D} = \frac{6}{3} = 2$$

$$D_2 = \begin{vmatrix} 1 & 5 \\ 2 & 1 \end{vmatrix} = 1 \cdot 1 - 5 \cdot 2 = -9$$

$$x_2 = \frac{D_2}{D} = \frac{-9}{3} = -3$$

Ответ: $x_1 = 2$, $x_2 = -3$ $\Rightarrow \begin{pmatrix} 2 \\ -3 \end{pmatrix}$

2.2.18

$$\begin{cases} x_1 - x_2 = 5 \\ 2x_1 + x_2 = 1 \end{cases}$$

□

1) $AX = B \Rightarrow X = A^{-1} \cdot B$

$$A = \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}$$

$$\det(A) = \begin{vmatrix} 1 & -1 \\ 2 & 1 \end{vmatrix} = 1 \cdot 1 - (-1) 2 = 3 \neq 0 \Rightarrow \exists A^{-1}$$

$$A^{-1} = \frac{1}{\det(A)} \cdot \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 1 & 1 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} \\ -\frac{2}{3} & \frac{1}{3} \end{pmatrix}$$

$$X = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} \\ -\frac{2}{3} & \frac{1}{3} \end{pmatrix} \cdot \begin{pmatrix} 5 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{3} \cdot 5 + \frac{1}{3} \cdot 1 \\ (-\frac{2}{3}) \cdot 5 + \frac{1}{3} \end{pmatrix} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$$

2) $D = \det(A) = 3$

$$D_1 = \begin{vmatrix} 5 & -1 \\ 1 & 1 \end{vmatrix} = 5 \cdot 1 - (-1) \cdot 1 = 6$$

$$x_1 = \frac{D_1}{D} = \frac{6}{3} = 2$$

$$D_2 = \begin{vmatrix} 1 & 5 \\ 2 & 1 \end{vmatrix} = 1 \cdot 1 - 5 \cdot 2 = -9$$

$$x_2 = \frac{D_2}{D} = \frac{-9}{3} = -3$$

Ответ: $x_1 = 2$, $x_2 = -3$ $\Rightarrow \begin{pmatrix} 2 \\ -3 \end{pmatrix}$

2.2.19

$$\begin{cases} x_1 - \sqrt{5}\, x_2 = 0 \\ 2\sqrt{5}\, x_1 - 5x_2 = -10 \end{cases}$$

1) $AX = B \Rightarrow X = A^{-1} \cdot B$

$$A = \begin{pmatrix} 1 & -\sqrt{5} \\ 2\sqrt{5} & -5 \end{pmatrix}$$

$$\det(A) = \begin{vmatrix} 1 & -\sqrt{5} \\ 2\sqrt{5} & -5 \end{vmatrix} = 1\cdot(-5) - (-\sqrt{5})\cdot 2\sqrt{5} = 5 \neq 0 \Rightarrow \exists A^{-1}$$

$$A^{-1} = \frac{1}{\det(A)} \cdot \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix} = \frac{1}{5} \cdot \begin{pmatrix} -5 & \sqrt{5} \\ -2\sqrt{5} & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{5}(-5) & \frac{1}{5}\sqrt{5} \\ \frac{1}{5}(-2\sqrt{5}) & \frac{1}{5}\cdot 1 \end{pmatrix} =$$

$$= \begin{pmatrix} -1 & \frac{\sqrt{5}}{5} \\ -\frac{2\sqrt{5}}{5} & \frac{1}{5} \end{pmatrix}$$

$$X = A^{-1} \cdot B = \begin{pmatrix} -1 & \frac{\sqrt{5}}{5} \\ -\frac{2\sqrt{5}}{5} & \frac{1}{5} \end{pmatrix} \cdot \begin{pmatrix} 0 \\ -10 \end{pmatrix} = \begin{pmatrix} (-1)\cdot 0 + \frac{\sqrt{5}}{5}\cdot(-10) \\ -\frac{2\sqrt{5}}{5}\cdot 0 + \frac{1}{5}\cdot(-10) \end{pmatrix} = \begin{pmatrix} -2\sqrt{5} \\ -2 \end{pmatrix}$$

2) $D = \det(A) = 5$

$$D_1 = \begin{vmatrix} 0 & -\sqrt{5} \\ -10 & -5 \end{vmatrix} = 0\cdot(-5) - (-\sqrt{5})\cdot(-10) = -10\sqrt{5}$$

$$x_1 = \frac{D_1}{D} = \frac{-10\sqrt{5}}{5} = -2\sqrt{5}$$

$$D_2 = \begin{vmatrix} 1 & 0 \\ 2\sqrt{5} & -10 \end{vmatrix} = 1\cdot(-10) - 0\cdot 2\sqrt{5} = -10$$

2.2.19

$$\begin{cases} x_1 - \sqrt{5}\,x_2 = 0 \\ 2\sqrt{5}\,x_1 - 5x_2 = -10 \end{cases}$$

1) $AX = B \Rightarrow X = A^{-1}\cdot B$

$$A = \begin{pmatrix} 1 & -\sqrt{5} \\ 2\sqrt{5} & -5 \end{pmatrix}$$

$$\det(A) = \begin{vmatrix} 1 & -\sqrt{5} \\ 2\sqrt{5} & -5 \end{vmatrix} = 1\cdot(-5) - (-\sqrt{5})\cdot 2\sqrt{5} = 5 \neq 0 \Rightarrow \exists A^{-1}$$

$$A^{-1} = \frac{1}{\det(A)} \cdot \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix} = \frac{1}{5}\cdot\begin{pmatrix} -5 & \sqrt{5} \\ -2\sqrt{5} & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{5}(-5) & \frac{1}{5}\sqrt{5} \\ \frac{1}{5}(-2\sqrt{5}) & \frac{1}{5}\cdot 1 \end{pmatrix} =$$

$$= \begin{pmatrix} -1 & \frac{\sqrt{5}}{5} \\ -\frac{2\sqrt{5}}{5} & \frac{1}{5} \end{pmatrix}$$

$$X = A^{-1}\cdot B = \begin{pmatrix} -1 & \frac{\sqrt{5}}{5} \\ -\frac{2\sqrt{5}}{5} & \frac{1}{5} \end{pmatrix} \cdot \begin{pmatrix} 0 \\ -10 \end{pmatrix} = \begin{pmatrix} (-1)\cdot 0 + \frac{\sqrt{5}}{5}\cdot(-10) \\ -\frac{2\sqrt{5}}{5}\cdot 0 + \frac{1}{5}\cdot(-10) \end{pmatrix} = \begin{pmatrix} -2\sqrt{5} \\ -2 \end{pmatrix}$$

2) $D = \det(A) = 5$

$$D_1 = \begin{vmatrix} 0 & -\sqrt{5} \\ -10 & -5 \end{vmatrix} = 0\cdot(-5) - (-\sqrt{5})\cdot(-10) = -10\sqrt{5}$$

$$x_1 = \frac{D_1}{D} = \frac{-10\sqrt{5}}{5} = -2\sqrt{5}$$

$$D_2 = \begin{vmatrix} 1 & 0 \\ 2\sqrt{5} & -10 \end{vmatrix} = 1\cdot(-10) - 0\cdot 2\sqrt{5} = -10$$

$$x_2 = \frac{D_2}{D} = \frac{-10}{5} = -2$$

Ответ: $x_1 = -2\sqrt{5}$ $\Rightarrow \begin{pmatrix} -2\sqrt{5} \\ -2 \end{pmatrix}$

$x_2 = -2$

2.2.20

$$\begin{cases} \alpha x - y = 2 \\ 2x + \alpha y = 1 \end{cases}$$

□

1) $AX = B \Rightarrow X = A^{-1} \cdot B$

$$\det(A) = \begin{vmatrix} \alpha & -1 \\ 2 & \alpha \end{vmatrix} = \alpha \cdot \alpha - (-1) \cdot 2 = \alpha^2 + 2 \neq 0 \Rightarrow \exists A^{-1}$$

$$A^{-1} = \frac{1}{\det(A)} \cdot \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix} = \begin{pmatrix} \frac{\alpha}{\alpha^2+2} & \frac{1}{\alpha^2+2} \\ \frac{-2}{\alpha+2} & \frac{\alpha}{\alpha^2+2} \end{pmatrix}$$

$$X = \begin{pmatrix} \frac{\alpha}{\alpha^2+2} & \frac{1}{\alpha^2+2} \\ \frac{-2}{\alpha^2+2} & \frac{\alpha}{\alpha^2+2} \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{2\alpha+1}{\alpha^2+2} \\ \frac{\alpha-4}{\alpha^2+2} \end{pmatrix}$$

2) $D = \det(A) = \alpha^2 + 2$

$$D_1 = \begin{vmatrix} 2 & -1 \\ 1 & \alpha \end{vmatrix} = 2 \cdot \alpha - (-1) \cdot 1 = 2\alpha + 1$$

$$x_1 = \frac{D_1}{D} = \frac{2\alpha+1}{\alpha^2+2}$$

$$D_2 = \begin{vmatrix} \alpha & 2 \\ 2 & 1 \end{vmatrix} = \alpha \cdot 1 - 2 \cdot 2 = \alpha - 4$$

$$x_2 = \frac{D_2}{D} = \frac{\alpha-4}{\alpha^2+2}$$

Ответ: $x_1 = \frac{2\alpha+1}{\alpha^2+2}$, $x_2 = \frac{\alpha-4}{\alpha^2+2}$ $\Rightarrow \begin{pmatrix} \frac{2\alpha+1}{\alpha^2+2} \\ \frac{\alpha-4}{\alpha^2+2} \end{pmatrix}$

2.2.23

$$\begin{cases} x_1 + 2x_2 + 3x_3 = 4 \\ 2x_1 + 6x_2 + 4x_3 = -6 \\ 3x_1 + 10x_2 + 8x_3 = -8 \end{cases}$$

Д

1) $AX = B \Rightarrow X = A^{-1} \cdot B$

$$\det(A) = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 6 & 4 \\ 3 & 10 & 8 \end{vmatrix} = 6 \neq 0 \Rightarrow \exists A^{-1}$$

$$A_{11} = \begin{vmatrix} 6 & 4 \\ 10 & 8 \end{vmatrix} = 8$$

$$A_{12} = -\begin{vmatrix} 2 & 4 \\ 3 & 8 \end{vmatrix} = -4$$

$$A_{13} = \begin{vmatrix} 2 & 6 \\ 3 & 10 \end{vmatrix} = 2$$

$$A_{21} = -\begin{vmatrix} 2 & 3 \\ 10 & 8 \end{vmatrix} = 14$$

$$A_{22} = \begin{vmatrix} 1 & 3 \\ 3 & 8 \end{vmatrix} = -1$$

$$A_{23} = -\begin{vmatrix} 1 & 2 \\ 3 & 10 \end{vmatrix} = -4$$

$$A_{31} = \begin{vmatrix} 2 & 3 \\ 6 & 4 \end{vmatrix} = -10$$

$$A_{32} = -\begin{vmatrix} 1 & 3 \\ 2 & 4 \end{vmatrix} = 2$$

$$A_{33} = \begin{vmatrix} 1 & 2 \\ 2 & 6 \end{vmatrix} = 2$$

$$\tilde{A} = \begin{pmatrix} 8 & -4 & 2 \\ 14 & -1 & -4 \\ -10 & 2 & 2 \end{pmatrix}^T = \begin{pmatrix} 8 & 14 & -10 \\ -4 & -1 & 2 \\ 2 & -4 & 2 \end{pmatrix}$$

$$A^{-1} = \frac{1}{\det(A)} \cdot \tilde{A} = \frac{1}{6} \begin{pmatrix} 8 & 14 & -10 \\ -4 & -1 & 2 \\ 2 & -4 & 2 \end{pmatrix} = \begin{pmatrix} \frac{4}{3} & \frac{7}{3} & -\frac{5}{3} \\ -\frac{2}{3} & -\frac{1}{6} & \frac{1}{3} \\ \frac{1}{3} & -\frac{2}{3} & \frac{1}{3} \end{pmatrix}$$

$$X = \begin{pmatrix} \frac{4}{3} & \frac{7}{3} & -\frac{5}{3} \\ -\frac{2}{3} & -\frac{1}{6} & \frac{1}{3} \\ \frac{1}{3} & -\frac{3}{3} & \frac{1}{3} \end{pmatrix} \cdot \begin{pmatrix} 4 \\ -6 \\ -8 \end{pmatrix} = \begin{pmatrix} \frac{14}{3} \\ -\frac{13}{3} \\ \frac{8}{3} \end{pmatrix}$$

$$D = \det(A) = 6$$

$$D_1 = \begin{vmatrix} 4 & 2 & 3 \\ -6 & 6 & 4 \\ -8 & 10 & 8 \end{vmatrix} \quad \det = 28$$

$$x_1 = \frac{D_1}{D} = \frac{28}{6} = \frac{14}{3}$$

$$D_2 = \begin{vmatrix} 1 & 4 & 3 \\ 2 & -6 & 4 \\ 3 & -8 & 8 \end{vmatrix} \cdot \det = -28$$

$$x_2 = -\frac{26}{6} = -\frac{13}{3}$$

$$D_3 = \begin{vmatrix} 1 & 2 & 4 \\ 2 & 6 & -6 \\ 3 & 10 & -8 \end{vmatrix} \quad \det = 16$$

$$x_3 = \frac{16}{6} = \frac{8}{3}$$

Ответ: $x_1 = \frac{14}{3}$

$x_2 = -\frac{13}{3} \Rightarrow \begin{pmatrix} \frac{14}{3} \\ -\frac{13}{3} \\ \frac{8}{3} \end{pmatrix}$

$x_3 = \frac{8}{3}$

2.2.24

$$\begin{cases} 3x+2y+z=1 \\ 6x+5y+4z=-2 \\ 9x+8y+7z=3 \end{cases}$$

$\square$

1) $AX=B \Rightarrow X=A^{-1}\cdot B$

$$\det(A)=\begin{vmatrix} 3 & 2 & 1 \\ 6 & 5 & 4 \\ 9 & 8 & 7 \end{vmatrix} \quad \det=0 \Rightarrow \nexists A^{-1}$$

Определитель равен 0 $\Rightarrow$ не решить

Ответ: не решить

2.2.26

$$\begin{cases} ax+by+z=1 \\ x+aby+z=b \\ x+by+az=1 \end{cases}$$

$\square$

1) $AX=B \Rightarrow X=A^{-1}\cdot B$

$$\det(A)=\begin{vmatrix} a & b & 1 \\ 1 & ab & 1 \\ 1 & b & a \end{vmatrix} = a\cdot ab\cdot a+b\cdot 1\cdot 1+b-ab-ba-ab=$$

$$=a^3b-3ab+2b=b(a+2)(a-1)^2$$

При $a^3b-3ab+2b=0$

$a=-2 \quad a=1 \quad b=0 \Rightarrow \nexists A^{-1} \Rightarrow$ не решить

При $a^3b-3ab+2b\neq 0$

$b\neq 0 \quad a\neq -2 \quad a\neq 1 \Rightarrow$

$$\Rightarrow A_{11} = \begin{vmatrix} ab & 1 \\ b & a \end{vmatrix} = a^2b - b$$

$$A_{12} = -\begin{vmatrix} 1 & 1 \\ 1 & a \end{vmatrix} = -a + 1$$

$$A_{13} = \begin{vmatrix} 1 & ab \\ 1 & b \end{vmatrix} = b - ab$$

$$A_{21} = -\begin{vmatrix} b & 1 \\ b & a \end{vmatrix} = b - ab$$

$$A_{22} = \begin{vmatrix} a & 1 \\ 1 & a \end{vmatrix} = a^2 - 1$$

$$A_{23} = -\begin{vmatrix} a & b \\ 1 & b \end{vmatrix} = b - ab$$

$$A_{31} = \begin{vmatrix} b & 1 \\ ab & 1 \end{vmatrix} = b - ab$$

$$A_{32} = -\begin{vmatrix} a & 1 \\ 1 & 1 \end{vmatrix} = 1 - a$$

$$A_{33} = \begin{vmatrix} a & b \\ 1 & ab \end{vmatrix} = a^2b - b$$

$$\tilde{A} = \begin{pmatrix} a^2b-b & 1-a & b-ab \\ b-ab & a^2-1 & b-ab \\ b-ab & 1-a & a^2b-b \end{pmatrix}^T = \begin{pmatrix} a^2b-b & b-ab & b-ab \\ 1-a & a^2-1 & 1-a \\ b-ab & b-ab & a^2b-b \end{pmatrix}$$

$$A^{-1} = \frac{1}{\det(A)} \cdot \tilde{A} = \frac{1}{a^3b-3ab+2b} \cdot \begin{pmatrix} a^2b-b & b-ab & b-ab \\ 1-a & a^2-1 & 1-a \\ b-ab & b-ab & a^2b-b \end{pmatrix}$$

$$= \begin{pmatrix} \frac{a^2b-b}{a^3b-3ab+2b} & \frac{b-ab}{a^3b-3ab+2b} & \frac{b-ab}{a^3b-3ab+2b} \\ \frac{1-a}{a^3b-3ab+2b} & \frac{a^2-1}{a^3b-3ab+2b} & \frac{1-a}{a^3b-3ab+2b} \\ \frac{b-ab}{a^3b-3ab+2b} & \frac{b-ab}{a^3b-3ab+2b} & \frac{a^2b-b}{a^3b-3ab+2b} \end{pmatrix} =$$

$$= \begin{pmatrix} \frac{a+1}{a^2+a-2} & \frac{-1}{a^2+a-2} & \frac{-1}{a^2+a-2} \\ \frac{-1}{a^2b+ab-2b} & \frac{a+1}{a^2b+ab-2b} & \frac{-1}{a^2b+ab-2b} \\ \frac{-1}{a^2+a-2} & \frac{1}{a^2+a-2} & \frac{a+1}{a^2+a-2} \end{pmatrix}$$

$$x = \qquad \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} =$$

$$= \begin{pmatrix} \dfrac{a-b}{a^2+a-2} \\ \\ \dfrac{ab+b-2}{a^2b+ab-2b} \\ \\ \dfrac{a-b}{a^2+a-2} \end{pmatrix}$$

2) $D = det(A) = a^3b - 3ab + 2b$

$$D_1 = \begin{vmatrix} 1 & b & 1 \\ b & ab & 1 \\ 1 & b & a \end{vmatrix} \quad det = a^2b + b^2 - ab - ab^2$$

$$x = \frac{D_1}{D} = \frac{a^2b + b^2 - ab - ab^2}{a^2b - 3ab + 2b} = \frac{a-b}{a^2+a-2}$$

$$D_2 = \begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & a \end{vmatrix} \overset{det}{=} a^2b - 2a - b + 2$$

$$y = \frac{D_2}{D} = \frac{a^2b - 2a - b + 2}{a^3b - 3ab + 2b} = \frac{ab+b-2}{a^2b + ab - 2b}$$

$$D_3 = \begin{vmatrix} a & b & 1 \\ 1 & ab & b \\ 1 & b & 1 \end{vmatrix} \overset{det}{=} a^2b + b^2 - ab - ab^2$$

$$z = \frac{D_3}{D} = \frac{a^2b + b^2 - ab - ab^2}{a^3b - 3ab + 2b} = \frac{a-b}{a^2+a-2}$$

Ответ: $x_1 = \frac{a-b}{a^2+a-2}$

$x_2 = \frac{ab+b-2}{a^2b+ab-2b} \Rightarrow \begin{pmatrix} \frac{a-b}{a^2+a-2} \\ \frac{ab+b-2}{a^2b+ab-2b} \\ \frac{a-b}{a^2+a-2} \end{pmatrix}$ $a \neq 1$, $a \neq -2$, $b \neq 0$

$x_3 = \frac{a-b}{a^2+a-2}$

2.2.27

$$\begin{cases} 2x_1 + x_2 + 4x_3 + 8x_4 = 0 \\ x_1 + 3x_2 - 6x_3 + 2x_4 = 0 \\ 3x_1 - 2x_2 + 2x_3 - 2x_4 = 0 \\ 2x_1 - x_2 + 2x_3 = 0 \end{cases}$$

$\square$

1) $AX = B \Rightarrow X = A^{-1} \cdot B$

$$\det(A) = \begin{vmatrix} 2 & 1 & 4 & 8 \\ 1 & 3 & -6 & 2 \\ 3 & -2 & 2 & -2 \\ 2 & -1 & 2 & 0 \end{vmatrix} = 24 \neq 0 \Rightarrow \exists A^{-1}$$

$$A^{-1} = \frac{1}{\det(A)} \cdot \begin{pmatrix} A_{11} & A_{12} & A_{13} & A_{14} \\ A_{21} & A_{22} & A_{23} & A_{24} \\ A_{31} & A_{32} & A_{33} & A_{34} \\ A_{41} & A_{42} & A_{43} & A_{44} \end{pmatrix}^T =$$

$$= \frac{1}{\det(A)} \cdot \begin{pmatrix} A_{11} & A_{21} & A_{31} & A_{41} \\ A_{12} & A_{22} & A_{32} & A_{42} \\ A_{13} & A_{23} & A_{33} & A_{43} \\ A_{14} & A_{24} & A_{34} & A_{44} \end{pmatrix}$$

$$X = A^{-1} \cdot B = \begin{pmatrix} A_{11} & A_{21} & A_{31} & A_{41} \\ A_{12} & A_{22} & A_{32} & A_{42} \\ A_{13} & A_{23} & A_{33} & A_{43} \\ A_{14} & A_{24} & A_{34} & A_{44} \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} =$$

$$= \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

2) $D = \det(A) = 24$

$$D_1 = \begin{vmatrix} 0 & 1 & 4 & 8 \\ 0 & 3 & -6 & 2 \\ 0 & -2 & 2 & -1 \\ 0 & -1 & 2 & 0 \end{vmatrix} = 0$$

$$x_1 = \frac{D_1}{D} = \frac{0}{24} = 0$$

$$D_2 = \begin{vmatrix} 2 & 0 & 4 & 8 \\ 1 & 0 & -6 & 2 \\ 3 & 0 & 2 & -2 \\ 2 & 0 & 2 & 0 \end{vmatrix} = 0$$

$$x_2 = \frac{D_2}{D} = \frac{0}{24} = 0$$

$$D_3 = \begin{vmatrix} 2 & 1 & 0 & 8 \\ 1 & 3 & 0 & 2 \\ 3 & -2 & 0 & -2 \\ 2 & -1 & 0 & 0 \end{vmatrix} = 0$$

$$x_3 = \frac{D_3}{D} = \frac{0}{24} = 0$$

$$D_4 = \begin{vmatrix} 2 & 1 & 4 & 0 \\ 1 & 3 & -6 & 0 \\ 3 & -2 & 2 & 0 \\ 2 & -1 & 2 & 0 \end{vmatrix} = 0$$

$$x_4 = \frac{D_4}{D} = \frac{0}{24} = 0$$

Ответ: $x_1 = 0$

$x_2 = 0$

$x_3 = 0$

$x_4 = 0$

$$\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

2.7.31

$f(x) = a \cdot \log_3 x + bx + c$

$\square$

$f(1) = 5: \quad a \cdot \log_3 1 + b \cdot 1 + c = 5$

$a \cdot 0 + b + c = 5$

$b + c = 5$

$f(3) = 8: \quad a \cdot \log_3 3 + b \cdot 3 + c = 8$

$a \cdot 1 + 3b + c = 8$

$a + 3b + c = 8$

$f(9) = 19: \quad a \cdot \log_3 9 + b \cdot 9 + c = 19$

$a \cdot 2 + 9b + c = 19$

$2a + 9b + c = 19$

$$\begin{cases} b + c = 5 \\ a + 3b + c = 8 \\ 2a + 9b + c = 19 \end{cases}$$

1) $Ax = B \Rightarrow x = A^{-1} \cdot B$

$$A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 3 & 1 \\ 2 & 9 & 1 \end{pmatrix}$$

$$\det(A) = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 3 & 1 \\ 2 & 9 & 1 \end{vmatrix} = 4 \neq 0 \Rightarrow \exists A^{-1}$$

$A_{11} = \begin{vmatrix} 3 & 1 \\ 2 & 1 \end{vmatrix} = -6$

$A_{12} = -\begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} = 1$

$A_{13} = \begin{vmatrix} 1 & 3 \\ 2 & 2 \end{vmatrix} = 3$

$A_{21} = -\begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} = 8$

$A_{22} = \begin{vmatrix} 0 & 1 \\ 2 & 1 \end{vmatrix} = -2$

$A_{23} = -\begin{vmatrix} 0 & 1 \\ 2 & 2 \end{vmatrix} = 2$

$A_{31} = \begin{vmatrix} 1 & 1 \\ 3 & 1 \end{vmatrix} = -2$

$A_{32} = -\begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = 1$

$A_{33} = \begin{vmatrix} 0 & 1 \\ 1 & 3 \end{vmatrix} = -1$

$$\tilde{A} = \begin{pmatrix} -6 & 1 & 3 \\ 8 & -2 & 2 \\ -2 & 1 & -1 \end{pmatrix}^T = \begin{pmatrix} -6 & 8 & -2 \\ 1 & -2 & 1 \\ 3 & 2 & -1 \end{pmatrix}$$

$$A^{-1} = \frac{1}{\det(A)} \cdot \tilde{A} = \frac{1}{4} \cdot \begin{pmatrix} -6 & 8 & -2 \\ 1 & -2 & 1 \\ 3 & 2 & -1 \end{pmatrix} =$$

$$= \begin{pmatrix} -\frac{3}{2} & 2 & -\frac{1}{2} \\ \frac{1}{4} & -\frac{1}{2} & \frac{1}{4} \\ \frac{3}{4} & \frac{1}{2} & -\frac{1}{4} \end{pmatrix}$$

$$x = \qquad \cdot \begin{pmatrix} 5 \\ 8 \\ 12 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix}$$

2) $D = \det A = 4$

$$D_1 = \begin{vmatrix} 5 & 1 & 1 \\ 8 & 3 & 1 \\ 12 & 9 & 1 \end{vmatrix} = -4$$

$$x_1 = \frac{D_1}{D} = \frac{-4}{4} = -1$$

$$D_2 = \begin{vmatrix} 0 & 5 & 1 \\ 1 & 8 & 1 \\ 2 & 12 & 1 \end{vmatrix} = 8$$

$$x_2 = \frac{D_2}{D} = \frac{8}{4} = 2$$

$$D_3 = \begin{vmatrix} 0 & 1 & 5 \\ 1 & 3 & 8 \\ 2 & 9 & 19 \end{vmatrix} = 12$$

$$x_3 = \frac{D_3}{D} = \frac{12}{4} = 3$$

Ответ: $x_1 = -1$

$x_2 = 2 \Rightarrow \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix}$

$x_3 = 3$

<u>№2.3.26</u>

$$\begin{cases} 2x_1 - x_2 = 0 \\ -4x_1 + 2x_2 = 0 \end{cases}$$

$\square$

$$(A|B) = \left(\begin{array}{cc|c} 2 & -1 & 0 \\ -4 & 2 & 0 \end{array}\right)_{II + I \cdot 2} \sim \left(\begin{array}{cc|c} 2 & -1 & 0 \\ 0 & 0 & 0 \end{array}\right)$$

$r(A) = r(A|B) = 1 \Rightarrow$ сов

~~неопр~~

$r(A) = 1$ и

$n - r = 2 - 1 =$ св

$|a_{11}| = |2| = 2 \neq 0 \Rightarrow$

x_1 – и

x_2 – св

$2x_1 - x_2 = 0$

$2x_1 = x_2$

$x_1 = \frac{1}{2} x_2$

$\exists\ x_2 = 2t \Rightarrow x_1 = t$

Оод $= (t; 2t)$

гсм $(t; 2t) = (1 \cdot t; 2t) = t(1; 2) = \{(1; 2)\}$

Ответ: Оод $(t; 2t)$

гсм $\{(1; 2)\}$

2.3.27

$$\begin{cases} x - \sqrt{3}y = 0 \\ \sqrt{3}x - 3y = 0 \end{cases}$$

$\square$

$$(A|B) = \left(\begin{array}{cc|c} 1 & -\sqrt{3} & 0 \\ \sqrt{3} & -3 & 0 \end{array}\right) \begin{matrix} \\ II - \sqrt{3} \cdot I \end{matrix} \sim \left(\begin{array}{cc|c} 1 & -\sqrt{3} & 0 \\ 0 & 0 & 0 \end{array}\right)$$

$r(A) = r(A|B) = 1 \Rightarrow$ совм

$n \geq r(A) = 1$ – глав

$n = 2 - 1 = 1$ – св

$|a_{11}| = |1| = 1 \neq 0 \Rightarrow$

$\Rightarrow$ БМ

$x_1 = \sqrt{3}\, x_2 = 0$

$x_1 = \sqrt{3}\, x_2$

] $x_2 = \frac{1}{\sqrt{3}} t \Rightarrow x_1 = \sqrt{3} \cdot \frac{1}{\sqrt{3}} t = t$

Ооу $\cdot (t; \frac{1}{\sqrt{3}} t)$

$(t; \frac{1}{\sqrt{3}} t) = t(1; \frac{1}{\sqrt{3}}) = \{(1; \frac{1}{\sqrt{3}})\}$

Ответ: Ооу $(t; \frac{1}{\sqrt{3}} t)$

racm $\{(1; \frac{1}{\sqrt{3}})\}$

2.3.28

$$\begin{cases} 3x + 4y = 0 \\ 4x - 3y = 0 \end{cases}$$

□

$$(A|B) = \left(\begin{array}{cc|c} 3 & 4 & 0 \\ 4 & -3 & 0 \end{array}\right)_{3 \cdot II - 4 \cdot I} \sim \left(\begin{array}{cc|c} 3 & 4 & 0 \\ 0 & -25 & 0 \end{array}\right)$$

$r(A) = r(A|B) = 2 \Rightarrow$ сов

$n = 2 \Rightarrow r = n$

$$\begin{cases} 3x + 4y = 0 \\ -25y = 0 \end{cases} \quad \begin{cases} x = 0 \\ y = 0 \end{cases}$$

Ооу $= (0; 0)$

Ответ: $(0; 0)$ - Ооу

2.3.35

$$\begin{cases} 2x_1 + x_2 - x_3 = 0 \\ x_1 - 2x_2 + x_3 = 0 \\ x_1 + 3x_2 - 2x_3 = 0 \\ x_1 + 8x_2 - 5x_3 = 0 \end{cases}$$

□

$$(A|B) = \left(\begin{array}{ccc|c} 2 & 1 & -1 & 0 \\ 1 & -2 & 1 & 0 \\ 1 & 3 & -2 & 0 \\ 1 & 8 & 5 & 0 \end{array}\right) \begin{array}{c} \\ 2\cdot II - I \\ 2\cdot III - I \\ 2\cdot IV - I \end{array} \sim \left(\begin{array}{ccc|c} 2 & 1 & -1 & 0 \\ 0 & -5 & 3 & 0 \\ 0 & 5 & -3 & 0 \\ 0 & 15 & 11 & 0 \end{array}\right) \begin{array}{c} \\ \\ III + II \\ IV + 3II \end{array} \sim$$

$$\sim \left(\begin{array}{ccc|c} 2 & 1 & -1 & 0 \\ 0 & -5 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 20 & 0 \end{array}\right) \underset{III \leftrightarrow IV}{\sim} \left(\begin{array}{ccc|c} 2 & 1 & -1 & 0 \\ 0 & -5 & 3 & 0 \\ 0 & 0 & 20 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$r(A) = r(A|B) = 3 \Rightarrow$ совм

$n = 3 \Rightarrow r = n$

$$\begin{cases} 2x_1 + x_2 - x_3 = 0 \\ -5x_2 + 3x_3 = 0 \\ 20x_3 = 0 \end{cases} \qquad \begin{cases} x_1 = 0 \\ x_2 = 0 \\ x_3 = 0 \end{cases}$$

Общ $= (0;0;0)$

Ответ: $(0;0;0)$

2.3.36

$$\begin{cases} x_1 - x_3 + x_5 = 0 \\ x_2 - x_4 + x_6 = 0 \\ x_1 - x_2 + x_5 - x_6 = 0 \\ x_2 - x_3 + x_6 = 0 \end{cases}$$

$\square$

$$(A|B)\begin{pmatrix} 1 & 0 & -1 & 0 & 1 & 0 & | & 0 \\ 0 & 1 & 0 & -1 & 0 & 1 & | & 0 \\ 1 & -1 & 0 & 0 & 1 & -1 & | & 0 \\ 0 & 1 & -1 & 0 & 0 & 1 & | & 0 \\ 1 & 0 & 0 & -1 & 1 & 0 & | & 0 \end{pmatrix} \underset{\substack{III-I \\ V-I}}{\sim} \begin{pmatrix} 1 & 0 & -1 & 0 & 1 & 0 & | & 0 \\ 0 & 1 & 0 & -1 & 0 & 1 & | & 0 \\ 0 & -1 & 1 & 0 & 0 & -1 & | & 0 \\ 0 & 1 & -1 & 0 & 0 & 1 & | & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & | & 0 \end{pmatrix}$$

$$\sim \begin{pmatrix} 1 & 0 & -1 & 0 & 1 & 0 & | & 0 \\ 0 & 1 & 0 & -1 & 0 & 1 & | & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & | & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & | & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & | & 0 \end{pmatrix} \underset{\substack{IV+III \\ V-III}}{\sim} \begin{pmatrix} 1 & 0 & -1 & 0 & 1 & 0 & | & 0 \\ 0 & 1 & 0 & -1 & 0 & 1 & | & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & | & 0 \end{pmatrix}$$

$r(A) = r(A|B) = 3 \Rightarrow$ совм

~~$r(A)$~~ $= 3$ н

$n - r = 6 - 3 = 3$ св

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1 \neq 0 \Rightarrow \begin{matrix} x_1, x_2, x_3 - и \\ x_4, x_5, x_6 - св \end{matrix}$$

$$\begin{cases} x_1 - x_3 + x_5 = 0 \\ x_2 - x_4 + x_6 = 0 \\ x_3 - x_4 = 0 \end{cases} \qquad \begin{cases} x_1 = x_4 - x_5 \\ x_2 = x_4 - x_6 \\ x_3 = x_4 \end{cases}$$

$$\text{]}\ \begin{matrix} x_4 = t_1 \\ x_5 = t_2 \\ x_6 = t_3 \end{matrix} \Rightarrow \begin{cases} x_1 = t_1 - t_2 \\ x_2 = t_1 - t_3 \\ x_3 = t_1 \end{cases}$$

Odz $(t_1 - t_2; t_1 - t_3; t_1; t_1; t_2; t_3) = 2\,\cancel{\text{ш}}$

$\cancel{t}$ $\{q\}$ =

$(t_1 - t_2; t_1 - t_3; t_1; t_2; t_3) = (1t_1; t_2 + 0t_3; t_1 - t_3 + 0t_1;$
$1t_1 + 0t_2 + 0t_3; 1t_1 + 0t_2 + 0t_2 + 0t_3; 0t_1 + t_2 + 0t_3; 0t_1$
$+ 0t_2 + t_3) = \{(1;1;1;1;0;0), (-1;0;0;0;1;0),$
$,(0;-1;0;0;0;1)\}$

Ответ: Odz $(t_1 - t_2; t_1 - t_3; t_1; t_1; t_2; t_3)$
базис $\{(1;1;1;1;0;0), (-1;0;0;0;1;0), (0;-1;0;$
$0;0;1)\}$

2.3.37

$$\begin{cases} x_1 + x_2 - x_3 + 2x_4 = 0 \\ x_1 + 3x_2 - 3x_3 + 4x_4 = 0 \\ 3x_1 + 2x_2 + x_3 = 0 \\ x_1 + 3x_2 - 5x_4 = 0 \end{cases}$$

$$(A|B) = \left(\begin{array}{cccc|c} 1 & 1 & -1 & 2 & 0 \\ 1 & 3 & -3 & 4 & 0 \\ 3 & 2 & 1 & 0 & 0 \\ 1 & 3 & 0 & -5 & 0 \end{array}\right) \begin{array}{c} \\ II - I \\ III - 3 \cdot I \\ IV - I \end{array} \sim \left(\begin{array}{cccc|c} 1 & 1 & -1 & 2 & 0 \\ 0 & 2 & -2 & 2 & 0 \\ 0 & -1 & 4 & -6 & 0 \\ 0 & 2 & 1 & -7 & 0 \end{array}\right) \sim$$

$$\sim \left(\begin{array}{cccc|c} 1 & 1 & -1 & 2 & 0 \\ 0 & 2 & -2 & 2 & 0 \\ 0 & 0 & 6 & -10 & 0 \\ 0 & 0 & 3 & -9 & 0 \end{array}\right) \sim \left(\begin{array}{cccc|c} 1 & 1 & -1 & 2 & 0 \\ 0 & 2 & -2 & 2 & 0 \\ 0 & 0 & 6 & -10 & 0 \\ 0 & 0 & 3 & -9 & 0 \end{array}\right) \sim_{2 \cdot IV - III} \left(\begin{array}{cccc|c} 1 & 1 & -1 & 2 & 0 \\ 0 & 2 & -2 & 2 & 0 \\ 0 & 0 & 6 & -10 & 0 \\ 0 & 0 & 0 & -8 & 0 \end{array}\right)$$

$r(A) = r(A|B) = 4 \Rightarrow$ совм

$n = 4 \Rightarrow r = n$

$$\begin{cases} x_1 + x_2 - x_3 + 2x_4 = 0 \\ 2x_2 - 2x_3 + 2x_4 = 0 \\ 6x_3 - 10x_4 = 0 \\ -8x_4 = 0 \end{cases} \qquad \begin{cases} x_1 = 0 \\ x_2 = 0 \\ x_3 = 0 \\ x_4 = 0 \end{cases}$$

Одн реш $(0;0;0;0)$

Ответ: Од $(0;0;0;0)$

2.3.41

$$\begin{cases} x_1 - 3x_2 + \ldots \\ 3x_1 + 2x_2 \ldots \\ 5x_1 + 6x_2 \ldots \\ 3x_1 + 9x_2 \ldots \end{cases}$$

$(A|B) = \ldots$

При ...

$r(A) = \ldots$

$r(A) \ldots$

$n - r = \ldots$

2.3.41

$$\begin{cases} x_1 - 3x_2 + x_3 - 2x_4 = 0 \\ 3x_1 + 2x_2 + 3x_4 = 0 \\ 5x_1 + 6x_2 - 4x_3 - x_4 = 0 \\ 3x_1 + 5x_2 - \lambda x_3 = 0 \end{cases}$$

$$(A|B) = \left(\begin{array}{cccc|c} 1 & -3 & 1 & 2 & 0 \\ 3 & 2 & 0 & 3 & 0 \\ 5 & 6 & -4 & -1 & 0 \\ 3 & 5 & -\lambda & 0 & 0 \end{array}\right) \begin{array}{c} \\ II-3\cdot I \\ III-5\cdot I \\ IV-3\cdot I \end{array} \sim \left(\begin{array}{cccc|c} 1 & -3 & 1 & 2 & 0 \\ 0 & 11 & -3 & -3 & 0 \\ 0 & 21 & -9 & -11 & 0 \\ 0 & 14 & -\lambda-3 & -6 & 0 \end{array}\right) \begin{array}{c} \\ \\ 11\cdot III-21\cdot II \\ 11\cdot IV-14\cdot II \end{array} \sim$$

$$\sim \left(\begin{array}{cccc|c} 1 & -3 & 1 & 2 & 0 \\ 0 & 11 & -3 & -3 & 0 \\ 0 & 0 & -36 & -58 & 0 \\ 0 & 0 & -11\lambda+9 & -24 & 0 \end{array}\right) \begin{array}{c} \\ \\ \\ 36\cdot IV+(-11\lambda+9)\cdot III \end{array} \sim \left(\begin{array}{cccc|c} 1 & -3 & 1 & 2 & 0 \\ 0 & 11 & -3 & -3 & 0 \\ 0 & 0 & -36 & -58 & 0 \\ 0 & 0 & 0 & -1386+638\lambda & 0 \end{array}\right)$$

При $-1386 + 638\lambda = 0$

$11(58\lambda - 126) = 0$

$58\lambda - 126 = 0$

$\lambda = \frac{126}{58}$

$r(A) = r(A|B) = 3 \Rightarrow$ совм

~~r(A)~~

$r(A) = 3 < 4$

$n - r = 1$ св

$$\begin{vmatrix} 1 & -3 & 1 \\ 0 & 11 & -3 \\ 0 & 0 & -36 \end{vmatrix} = -396 \neq 0 \Rightarrow x_1; x_2, x_3 \Rightarrow [illegible]$$

$x_4 \Rightarrow$ СВ

$$\begin{cases} x_1 - 3x_2 + x_3 + 2x_4 = 0 \\ 11x_2 - 3x_3 - 3x_4 = 0 \\ -36x_3 - 58x_4 = 0 \end{cases} \qquad \begin{cases} x_1 = -\frac{173}{99} x_4 \\ x_2 = -\frac{47}{66} x_4 \\ x_3 = -\frac{29}{18} x_4 \end{cases}$$

1) $x_1 - 3\left(-\frac{47}{66}\right)x_4 + \left(-\frac{29}{18}\right)x_4 + 2x_4 = 0$

$x_1 = -\frac{173}{99} x_4$

2) $11x_2 - 3\cdot\left(-\frac{29}{18}\right)x_4 - 3x_4 = 0$

$11x_2 + \frac{47}{6}x_4 = 0$

$x_2 = -\frac{47}{66}x_4$

3) $-36x_3 = 58x_4$

$x_3 = -\frac{58}{36}x_4$

$x_3 = -\frac{29}{18}x_4$

] $x_4 = t$

Общ реш $\left(-\frac{173}{99}t; -\frac{47}{66}t; -\frac{29}{18}t; t\right)$

Част $\left(-\frac{173}{99}; -\frac{47}{66}; -\frac{29}{18}; 1\right)$

2) при $98\lambda - 126 \neq 0 \Rightarrow \lambda \neq \frac{126}{98}$

$r(A) = r(A|B) = 4$ совм

$n = 4 \Rightarrow r = n$

$$\begin{cases} x_1 - 3x_2 + x_3 + 2x_4 = 0 \\ 11x_2 - 3x_3 - 3x = 0 \\ -36x_3 - 58x_4 = 0 \\ (1638\lambda - 1386)x_4 = 0 \end{cases} \qquad \begin{cases} x_1 = 0 \\ x_2 = 0 \\ x_3 = 0 \\ x_4 = 0 \end{cases}$$

Отв: $(0;0;0;0)$

Ответ: 1) Отв $\left(-\frac{173}{99}t;\ \frac{47}{66}t;\ -\frac{29}{18}t;\ t\right)$

$\left\{\left(-\frac{173}{99};\ \frac{47}{66};\ -\frac{29}{18};\ 1\right)\right\}$

и и

2) при $\lambda \neq \frac{126}{58}$: отв $(0;0;0;0)$